

IEEE Transactions on Multimedia

Decision Letter (MM-024883)**From:** shiqwang@cityu.edu.hk**To:** zqshen@ntu.edu.sg**CC:** ygwen@ntu.edu.sg, shiqwang@cityu.edu.hk**Subject:** Decision: MAJOR REVISION (RQ) - MM-024883, EGRA: Toward Enhanced Behavior Graphs and Representation Alignment for Multimodal Recommendation**Body:** 01-Jan-2026

Dr Zhiqi Shen
 Nanyang Technological University
 Singapore
 Singapore
 Singapore

Paper:MM-024883 EGRA: Toward Enhanced Behavior Graphs and Representation Alignment for Multimodal Recommendation

Dear Dr Shen,

I am writing to you concerning the above referenced manuscript, which you submitted to the IEEE Transactions on Multimedia. (**See note below about attachments).

Based on the enclosed set of reviews, it was recommended that the manuscript be REVISED AND RESUBMITTED (RQ).

When you are ready to submit your revision, visit the following link:
[https://iee.atyponrex.com/submission/submissionBoard/REX-PROD-2-1F6D9219-45FC-46A6-8FDE-A61DB153EBA1-23F6CE66-BCA2-4BB3-AE33-788D8BDA7E33-44629/current?](https://iee.atyponrex.com/submission/submissionBoard/REX-PROD-2-1F6D9219-45FC-46A6-8FDE-A61DB153EBA1-23F6CE66-BCA2-4BB3-AE33-788D8BDA7E33-44629/current?doctype=external)
 idtype=external

* If you have any questions regarding the reviews, please contact the Associate Editor who managed your paper. All other inquiries should be directed to the Admin.

Best regards,
 Professor Shiqi Wang
 Associate Editor, IEEE Transactions on Multimedia
 shiqwang@cityu.edu.hk

Reviewer Comments:
 Reviewer: 1

Recommendation: RQ - Review Again After Major Changes

Comments:

This paper proposes a multimodal recommendation framework named EGRA, which enhances recommendation performance by augmenting behavior graphs and improving representation alignment mechanisms. Extensive experiments were conducted on five public datasets, demonstrating significant improvements over existing SOTA methods. The overall work is solid, with thorough experimentation and a certain degree of innovation and practicality. However, there are still some aspects that could be improved.

Weak points:

- 1.The idea of using pre-trained representations to construct enhanced semantic graphs has precedents in existing literature. As one of the key contributions of this paper, the authors should clarify how their approach differs from existing methods.
- 2.Although the dual-level alignment mechanism is carefully designed, it essentially remains an extension of weight adjustment in contrastive learning, with limited theoretical innovation.
- 3.The alignment loss function (Eq. 17) is complex in design, and there is a lack of detailed analysis and explanation regarding its effectiveness.
- 4.No comparison of training or inference time is provided. Although computational complexity analysis is included, there is no validation of actual runtime efficiency.
- 5.The baseline method GUME also performs modality-based graph enhancement. While the paper compares with GUME and demonstrates superiority, the core difference mainly lies in "using pre-trained embeddings (MGCN) instead of raw features," which appears more like an engineering improvement rather than a fundamental algorithmic or theoretical breakthrough. The authors are advised to address this point to clarify the innovation of their work.

Additional Questions:

1. Is the topic appropriate for publication in these transactions?: Excellent Match

1. Is the paper technically sound?: Yes

2. How would you rate the technical novelty of the paper?: Novel Enough for Publication

Explain: Strong points:

1. Clearly identifies the shortcomings of existing methods in terms of behavior graph sparsity and representation misalignment, and proposes targeted solutions.
 2. Uses pre-trained models to generate item representations for graph construction, leveraging multimodal information while avoiding noise interference; the dual-level alignment mechanism is meticulously designed and aligns well with training dynamics.
 3. Comprehensive experiments were conducted on five datasets, including conventional evaluation, long-tail evaluation, ablation studies, and parameter analysis, yielding convincing results.

3. Is the contribution significant?: Moderately Significant

4. Is the coverage of the topic sufficiently comprehensive and balanced?: Yes

5. Rate the Bibliography: Satisfactory

1. How would you rate the overall organization of the paper?: Satisfactory

2. Are the title and abstract satisfactory?: Yes

3. Is the length of the paper appropriate? If not, recommend what should be added or eliminated.: Yes

4. Are symbols, terms, and concepts adequately defined?: Yes

5. How do you rate the English usage?: Satisfactory

If you are suggesting additional references they must be entered in the text box provided. All suggestions must include full bibliographic information plus a DOI.

If you are not suggesting any references, type N/A.: N/A

Reviewer: 2

Recommendation: RQ - Review Again After Major Changes

Comments:

This paper investigates the multimodal recommendation problem and proposes an EGRA system that leverages both visual and textual information. The key contributions lie in (a) effectively leveraging pretrained representations and (b) introducing a dynamic weighting mechanism for fusion. Experiments demonstrate that the proposed method achieves state-of-the-art performance. Overall, the framework is well-designed and supported by convincing experiments that show consistent improvements across multiple datasets. However, the following issues should be addressed before publication:

- Since the approach relies on pretrained representations, both theoretical analysis and experimental evaluation regarding computational complexity are critically needed to provide a comprehensive understanding of the system's efficiency. The statement in Table I appears too simple without supporting analysis, and the complexity evaluation in the experiment part is missing.
- It would be highly insightful if the authors could briefly explore the potential performance gains achievable by replacing the current pretrained models with more advanced or higher-capacity ones.
- Let us confirm that this paper is not the first one to effectively introduce the pretrained representatives. If this is the case, it is necessary to explicitly indicate the differences between this work and prior approaches in the introduction and related work sections. This will help better situate and highlight the specific contribution of this study.
- Please supplement the missing results for the other two datasets in Table III.
- Minor revision: There appears to be an inconsistency between the text in Section V-D, which states "into three representative backbone models: LightGCN, BM3, and MGCN," and Table IV, which includes LATTICE.

Additional Questions:

1. Is the topic appropriate for publication in these transactions?: Excellent Match

1. Is the paper technically sound?: Yes

2. How would you rate the technical novelty of the paper?: Novel Enough for Publication

Explain: It delivers sound designs and successfully achieves the SOTA performance.

3. Is the contribution significant?: Significant
4. Is the coverage of the topic sufficiently comprehensive and balanced?: Yes
5. Rate the Bibliography: Satisfactory
1. How would you rate the overall organization of the paper?: Satisfactory
2. Are the title and abstract satisfactory?: Yes
3. Is the length of the paper appropriate? If not, recommend what should be added or eliminated.: Yes
4. Are symbols, terms, and concepts adequately defined?: Yes
5. How do you rate the English usage?: Satisfactory

If you are suggesting additional references they must be entered in the text box provided. All suggestions must include full bibliographic information plus a DOI.

If you are not suggesting any references, type N/A.: N/A

Reviewer: 3

Recommendation: RQ - Review Again After Major Changes

Comments:

The paper proposes a framework called EGRA (Enhanced Behavior Graphs and Representation Alignment) for Multimodal Recommendation. It aims to address two limitations in existing MMR methods: the noise in raw modality features when constructing item graphs, and the lack of dynamic alignment between modality and behavior representations. To solve these, EGRA introduces an enhanced behavior graph using representations from a pre-trained MMR model (MGCN) instead of raw features. It also proposes a Bi-Level Dynamic Alignment Weighting mechanism (entity-wise and epoch-wise) to balance the learning process. Experiments on five datasets demonstrate improvements over baselines.

I have the following issues:

1. Proofreading is necessary to improve the quality of the manuscript. For example, in Section IV-C, paragraph "Epoch-Wise Dynamic Weighting", the sentence ends with double periods. Moreover, the paper frequently uses "Besides," to start paragraphs.
2. The method's success seems heavily dependent on the quality of the pre-trained MGCN. If MGCN fails to capture good semantics on a specific dataset, will EGRA also fail?
3. Using a pre-trained model to initialize or guide a graph is a known technique (e.g., in knowledge distillation or graph structure learning). The authors need to clarify why using MGCN specifically constitutes a significant technical innovation rather than just an engineering combination. Have you tried other backbones for pre-training? If EGRA only works with MGCN, it limits the framework's generality.
4. In the Related Work, the authors mention LLM-based recommendation methods like LLM-ESR and Llama4Rec. However, these are not included in the main performance comparison. Comparing with LLM-enhanced representation methods would be very relevant to prove the superiority of the proposed graph enhancement. If they are too computationally expensive, at least a discussion or a smaller-scale comparison is needed.
5. It would be convincing to see a distribution of weights for "Head" vs "Tail" items to verify if the model truly assigns higher weights to tail items as claimed.
6. The paper states: "The enhanced behavior graph is pre-constructed before training begins and thus not included in the runtime complexity estimation". In real-world applications, one cannot simply ignore the pre-training phase. A more detailed clarification is needed at this point.

Additional Questions:

1. Is the topic appropriate for publication in these transactions?: Excellent Match
1. Is the paper technically sound?: Yes
2. How would you rate the technical novelty of the paper?: Novel Enough for Publication

Explain: The paper proposes a framework called EGRA (Enhanced Behavior Graphs and Representation Alignment) for Multimodal Recommendation. It aims to address two limitations in existing MMR methods: the noise in raw modality features when constructing item graphs, and the lack of dynamic alignment between modality and behavior representations. To solve these, EGRA introduces an enhanced behavior graph using representations from a pre-trained MMR model (MGCN) instead of raw features. It also proposes a Bi-Level Dynamic Alignment Weighting mechanism (entity-wise and epoch-wise) to balance the learning process. Experiments on five datasets demonstrate improvements over baselines.

3. Is the contribution significant?: Moderately Significant
4. Is the coverage of the topic sufficiently comprehensive and balanced?: Yes

5. Rate the Bibliography: Satisfactory

1. How would you rate the overall organization of the paper?: Could be improved
2. Are the title and abstract satisfactory?: Yes
3. Is the length of the paper appropriate? If not, recommend what should be added or eliminated.: Yes
4. Are symbols, terms, and concepts adequately defined?: Not always
5. How do you rate the English usage?: Needs improvement

If you are suggesting additional references they must be entered in the text box provided. All suggestions must include full bibliographic information plus a DOI.

If you are not suggesting any references, type N/A.: N/A

Date Sent: 01-Jan-2026

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